TECHNICAL REPORT

FastSpeech 2-Based Text-to-Speech Synthesis
English Implementation

Language	English
Dataset	Svarah
Reference	FastSpeech 2 GitHub implementation, adapted
Training run	5,000-step pilot
Scope	English only; Telugu excluded from this report
Assignment deadline	7 November 2026

Prepared from the English implementation workflow, configuration, logs, feature audits, and synthesis outputs recorded during development.

Executive Summary

This report documents the English-language adaptation of a reference FastSpeech 2 implementation using the Svarah speech dataset. The work covered dataset selection, forced alignment with Montreal Forced Aligner (MFA), phoneme-level feature extraction, conversion to the repository's preprocessing format, configuration adaptation, pilot training, inference, and preliminary variance-adaptor analysis.

The selected subset contains approximately 5.006 hours of training audio and 1.004 hours of validation audio. From 3,461 selected recordings, 3,445 produced valid FastSpeech 2 feature files (99.54% coverage). A 5,000-step pilot completed and saved checkpoints at steps 1,000–5,000. Final recorded training total loss was 4.3053 and validation total loss was 4.5679.

Important scope qualification: the 5,000-step run is a pilot, not evidence of convergence. The five preliminary synthesis sentences used during development are different from the exact test sentences in the assignment PDF. The report records this gap; final assignment-compliant test audio and variance plots remain to be generated.

1. Implementation Approach and Architecture

The reference implementation was used as the base. FastSpeech 2 is a non-autoregressive text-to-mel model. Its processing path comprises a phoneme frontend, Transformer encoder, variance adaptor, length regulator, Transformer decoder, and postnet. The variance adaptor predicts duration, pitch, and energy. The length regulator expands phoneme representations according to predicted durations to create a mel-frame sequence. A separate HiFi-GAN vocoder converts the mel spectrogram into waveform audio.

Architecture components retained

- **Text frontend:** repository symbol inventory and text-to-sequence conversion using English ARPAbet phones represented with @-prefixed symbols.
- **Encoder:** 4 Transformer layers, 2 attention heads, hidden size 256.
- **Variance adaptor:** duration, pitch, and energy predictors; pitch and energy configured at phoneme level; length regulator for duration-based expansion.
- **Decoder and postnet:** 6 Transformer decoder layers, 2 heads, hidden size 256, followed by the mel postnet.
- **Vocoder:** pretrained HiFi-GAN LJSpeech generator. This is separate from the trained FastSpeech 2 acoustic-model checkpoints.

English-specific adaptations

- Created independent **config/Svarah/** preprocessing, model, and training YAML files; original LJSpeech configurations were retained.
- Configured the English frontend with **english_cleaners** and an MFA-derived English US ARPAbet pronunciation dictionary.
- Modified **synthesize.py** lexicon reading to detect and strip four numeric MFA metadata fields while preserving ordinary dictionary entries.
- Updated checkpoint loading in **utils/model.py** to use **torch.load(..., map_location=device, weights_only=False)** for the trusted checkpoint generated by this run, whose metadata was incompatible with the PyTorch 2.6+ weights-only default.
- Converted NPZ features into the repository's expected directory and metadata format. The original Dataset and training path passed compatibility tests.
- Used the logical speaker label **speaker_01** because speaker IDs were not supplied in the prepared Svarah manifest. This is a formatting convention, not proof that the source recordings contain only one actual speaker.

2. Dataset Selection, Subsampling, and Preprocessing

Svarah was selected for English. The prepared manifests contain 2,908 training and 553 validation records, selected to meet the requested approximate five-hour/one-hour duration targets. Training and validation recording IDs were checked; overlap was zero. The exact ranking/randomization rule used during selection is not preserved in the run notes and should be documented from the original selection script if available.

Split	Manifest records	Approx. duration	Valid feature files
Training	2,908	5.006 hours	2,897

Split	Manifest records	Approx. duration	Valid feature files
Validation	553	1.004 hours	548
Total	3,461	6.010 hours	3,445

Alignment and pronunciation preparation

Audio and transcripts were organized into MFA corpus folders as WAV/LAB pairs. MFA used the English US ARPA acoustic model (version 3.4.3) in single-speaker mode and a preserved pronunciation dictionary. The final dictionary contained 200,960 unique words and 210,387 pronunciation entries. Train/validation vocabulary coverage was reported as complete after number normalization and G2P handling of alphabetic OOV items.

MFA aligned 2,900/2,908 training records and 548/553 validation records: 3,448/3,461 overall (99.62%). Eight training and five validation records lacked alignments. Three additional training records failed pitch interpolation. Final feature output therefore contains 2,897 training and 548 validation recordings.

Feature extraction and conversion

The MFA phone tier was used to derive phoneme labels and durations. Empty intervals were ignored; gaps greater than 1 ms were represented as @sil. Explicit leading/trailing silence was trimmed by the alignment helper. Audio was resampled to 22,050 Hz. Mel features were calculated with the repository TacotronSTFT implementation. Pitch used PyWorld DIO followed by StoneMask. Pitch and energy were phoneme-level features; mel frames were cropped to the sum of phoneme durations.

Feature configuration: sample rate 22,050 Hz; FFT/filter length 1,024; window length 1,024; hop length 256; 80 mel channels; mel frequency range 0–8,000 Hz. Feature checks covered dimensions, finite values, phone-symbol validity, and agreement between duration sums and mel-frame counts.

NPZ records were converted into mel/, pitch/, energy/, and duration/ folders, with train.txt, val.txt, speakers.json, and stats.json. Mel arrays use shape (T, 80); pitch, energy, and duration arrays are phoneme-length. Metadata follows basename|speaker|phone_sequence|raw_text. The actual repository Dataset loaded a converted sample successfully, and full checks passed for all 3,445 feature records.

Pitch and energy normalization statistics were fitted on training data only: pitch mean 177.8983 and standard deviation 56.8179; energy mean 27.3366 and standard deviation 26.1940. The model reads these saved statistics and uses linear quantization bins for the pitch and energy embeddings.

3. Model Configuration and Training

Setting	English configuration
Encoder / decoder	4 layers / 6 layers; 2 heads; hidden size 256
Convolution	Filter size 1,024; kernel [9, 1]
Variance predictors	Filter size 256; kernel 3; dropout 0.5
Pitch / energy	Phoneme-level; linear quantization; 256 bins
Optimizer	Adam; betas 0.9/0.98; epsilon 1e-9
Batch size / gradient clipping	16 / 1.0
Pilot warm-up / total steps	1,000 / 5,000
Validation / checkpoint interval	1,000 / 1,000 steps

The original configuration described a substantially longer run. For this pilot, total_step was set to 5,000, warm_up_step to 1,000, anneal_steps to an empty list, and logging, validation, and checkpoint intervals were shortened. The run completed at step 5,000 and saved checkpoints at 1,000, 2,000, 3,000, 4,000, and 5,000. Each checkpoint was approximately 398.73 MB. The model contained approximately 35.16 million parameters. GPU execution was confirmed through CUDA tensor, forward, backward, and optimizer checks.

4. Training and Validation Loss Analysis

Step	Training total loss	Validation total loss
100	≈ 7.75	Not recorded
1,000	4.0164	4.2303
5,000	4.3053	4.5679

At step 5,000, training component losses were mel 1.0277, mel postnet 1.0256, pitch 0.9991, energy 0.8500, and duration 0.4028. Final validation component losses were mel 1.0765, mel postnet 1.0775, pitch 1.0802, energy 0.9660, and duration 0.3677.

The early decrease from approximately 7.75 at step 100 to 4.0164 at step 1,000 indicates learning during the pilot. However, validation total loss at step 5,000 (4.5679) was higher than at step 1,000 (4.2303). The available evidence therefore does not demonstrate continued validation improvement or convergence. There are only five validation observations (steps 1,000–5,000), which is insufficient to claim definitive overfitting. The curves should be presented as pilot diagnostics, not proof of a converged model.

5. Speech Generation and Audio Observations

The pretrained HiFi-GAN LJSpeech generator loaded successfully on CUDA and passed a mel-to-waveform smoke test. FastSpeech 2 inference using checkpoint 5,000 also completed. Five preliminary sentences were synthesized as 22,050 Hz WAV files, with durations approximately 1.39, 3.67, 2.12, 2.45, and 3.91 seconds.

The user listened to the preliminary sample “The sun rises in the east.” and reported that it sounded good. Listening quality for the other four preliminary samples was not confirmed in the recorded workflow. No MOS, intelligibility score, pronunciation error rate, or speaker-similarity evaluation was performed; consequently, no broader quality claim is made.

Assignment test-set discrepancy

The assignment specifies four English IDs, with E2 containing two variants (a) and (b), giving five required sentence variants. The five preliminary sentences used in development are different. Those WAVs are smoke-test outputs, not final assignment-compliant samples. The exact prescribed text must be preserved for final synthesis.

- **E1:** The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.
- **E2(a):** The orchestra's conduct during the ceremony was praised by the audience.
- **E2(b):** The researchers will conduct an experiment to evaluate the proposed method.
- **E3:** The rhyth m of the orchestra changed unexpectedly. (Spacing shown as in the supplied assignment.)
- **E4:** The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

6. Variance-Adaptor Prediction Analysis

For each preliminary sentence, model inference returned phoneme-level pitch and energy predictions, log-duration predictions, and rounded duration predictions. Duration was converted from mel frames to seconds using hop_length/sample_rate = 256/22,050 ≈ 0.01161 seconds per mel frame. Pitch and energy were converted from normalized space using the training statistics in stats.json.

The preliminary output contains five duration plots, five pitch plots, five energy plots, and five CSV files. Numerical audits found finite predictions for every sample and no zero-duration phonemes. Predicted total durations were 1.393 s (E1), 3.669 s (E2), 2.125 s (E3), 2.450 s (E4), and 3.913 s (E5).

Duration predictions were concentrated: per-sentence mean was approximately 7.02–7.06 frames per phoneme, standard deviation approximately 0.32–0.54 frames, and only three distinct rounded duration values occurred in each sentence. This may indicate that the pilot predictor has not learned the full variation of phoneme durations. A direct comparison with the training duration-target distribution is needed before diagnosing underfitting.

Pitch predictions were not constant: standard deviations ranged from approximately 6.43 to 10.51 Hz. Energy predictions were also non-constant, with standard deviations approximately 2.96–4.91 in original feature units. These values show some phoneme-level variation, but their relatively narrow ranges require comparison with training-target distributions and listening-based prosody evaluation. A pitch prediction at every phoneme should not automatically be treated as reliable measured F0 for unvoiced phonemes.

The existing plots correspond to preliminary development sentences, not the assignment's prescribed samples. Final analysis must use the required E1, E2(a), E2(b), E3, and E4 outputs. Discuss duration in relation to phoneme identity and rhythm, pitch/F0 in relation to intonation and voicing, and energy in relation to emphasis and stress.

7. Limitations and Possible Improvements

- **Short pilot:** 5,000 steps do not establish convergence. Continue training under controlled validation monitoring and compare checkpoints using both losses and listening tests.
- **Variance predictor concentration:** compare predicted and target distributions; investigate predictor learning if predictions remain substantially narrower.
- **Exact test sentences:** regenerate the five required variants before final evaluation.
- **Frontend robustness:** verify long words, contractions, homographs such as “conduct,” number/time normalization, and unusual spacing in E3.
- **Audio evaluation:** listen to every required sample and document pronunciation, intelligibility, prosody, clipping, and artifacts; optionally collect human ratings.
- **Vocoder/domain mismatch:** HiFi-GAN LJSpeech is pretrained on a different corpus. Evaluate timbre/naturalness and consider compatible vocoder adaptation if resources allow.
- **Reproducibility:** capture package versions, runtime/hardware, dataset manifest hashes, code commit, and final configs.

8. Reproducibility: Environment, Paths, and Commands

The recorded development environment was Google Colab with a CUDA-enabled PyTorch runtime, Python 3.12.6, and the cloned repository at /content/FastSpeech2. The workflow used micromamba/MFA for forced alignment and PyWorld for pitch extraction. A complete package freeze was not captured in the retained notes; generate and commit requirements.txt from the final working runtime rather than guessing dependency versions.

Configuration: config/Svarah/preprocess.yaml, config/Svarah/model.yaml, config/Svarah/train.yaml
Preprocessed data: /content/drive/MyDrive/FastSpeech2_Assignment/data/processed/english/fastspeech2_preprocessed
Checkpoints: /content/drive/MyDrive/FastSpeech2_Assignment/output/ckpt/Svarah
Results: /content/drive/MyDrive/FastSpeech2_Assignment/Results/English

Pilot training command

```
%cd /content/FastSpeech2
python train.py --restore_step 0 --preprocess_config config/Svarah/preprocess.yaml --model_config
config/Svarah/model.yaml --train_config config/Svarah/train.yaml
```

Single-sentence inference command used during development

```
%cd /content/FastSpeech2
python synthesize.py --restore_step 5000 --mode single --text "The sun rises in the east." -p
config/Svarah/preprocess.yaml -m config/Svarah/model.yaml -t config/Svarah/train.yaml
```

This inference command documents the invocation pattern only. Substitute each exact assignment sentence and keep E2(a) and E2(b) as separate outputs. The final repository should also include the exact preprocessing/alignment commands from the scripts or notebook used to create the dataset.

9. English Deliverable Status

Deliverable	Status	Notes
Dataset preparation	Completed	Approx. 5.006 h train / 1.004 h validation; zero ID overlap.
Alignment/features	Completed	3,445 valid feature records; 99.54% coverage.
Configs/data compatibility	Completed	Repository Dataset and model forward/backward checks passed.
Pilot training/checkpoints	Completed	5,000-step pilot; convergence not established.
Loss plots	Generated	50 training points and 5 validation points.
Preliminary audio	Generated	Five development sentences; one listening confirmation recorded.

Deliverable	Status	Notes
Variance plots	Generated	15 preliminary plots and five CSV files.
Exact assignment audio/plots	Pending	Regenerate E1, E2(a), E2(b), E3, E4 exactly.
Final audio/variance analysis	Pending	Requires listening review and required-sentence plots.
Dependency lockfile/final code package	Pending	Capture exact package versions and finalize repository.

10. Conclusion

The English FastSpeech 2 pipeline has been adapted to Svarah and exercised through feature preparation, repository-compatible loading, model training, checkpoint saving, inference, and preliminary variance-adaptor visualization. The feature pipeline achieved 99.54% coverage of selected recordings, and the 5,000-step pilot showed a substantial early reduction in training loss. These results establish a functional implementation milestone, but do not establish convergence or final speech quality. The English work should be considered a pilot until the exact assignment sentences are synthesized, listened to, analyzed, and packaged.

Source and scope note

This report is based on the English project configuration, logs, validation outputs, feature audits, synthesis checks, and assignment specification available during development. It intentionally excludes Telugu. Unavailable information, including a complete dependency lockfile and formal audio-quality scores, is identified rather than estimated.